ASSIGNMENT -1
MODULE 1

COMPUTER NETWORKS
(PCCST501)

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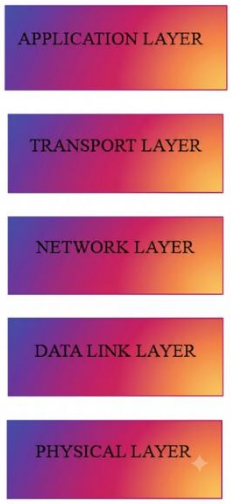
How the Internet Delivers a Web Page

Website Chosen: Instagram (https://www.instagram.com)

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer

The TCP/IP protocol suite is a five-layer architecture that enables communication between devices over the Internet. Each layer performs a specific function and provides services to the layer above it.



Application Layer

The Application Layer provides services directly to users and applications. It uses protocols such as **HTTP, HTTPS, DNS, FTP, SMTP, and SSH**. When a user enters ***www.instagram.com***, the browser uses **HTTPS** to securely request Instagram's webpage and receives photos, videos, reels, and other content from the server.

Transport Layer

The Transport Layer provides **process-to-process communication**. It divides data into **segments** and ensures reliable delivery using **TCP**. This allows Instagram's photos, reels, messages, and webpages to reach the user correctly and in order.

Network Layer

The Network Layer uses the **Internet Protocol (IP)** to route data packets from the user's device to Instagram's servers through different networks until they reach the destination.

Data Link Layer

The Data Link Layer transfers **frames** between devices on the same network. It uses **MAC addresses** and prepares data for transmission over Wi-Fi or Ethernet.

Physical Layer

The Physical Layer transmits data as **electrical, optical, or wireless signals** through cables, fiber optics, or Wi-Fi, enabling communication between devices.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address

When a user enters www.instagram.com into the browser, the browser cannot communicate using the domain name alone because communication on the Internet requires an IP address. The browser first contacts the Domain Name System (DNS). DNS is an application-layer protocol that translates a human-readable domain name into the corresponding IP address. After receiving Instagram's IP address, the browser knows the location of Instagram's web server and establishes communication with it. DNS acts as a distributed database that maps domain names to IP addresses. Without DNS, users would have to remember numerical IP addresses instead of easy-to-read website names.

www.instagram.com

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▼

DNS Server

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▼

Instagram Server IP Address

3. Describe How HTTP Is Used to Request and Receive the Webpage

HTTP (HyperText Transfer Protocol) is used to access web pages. Instagram uses **HTTPS**, the secure version of HTTP, to protect communication between the browser and the server.

The communication takes place as follows:

1. The user enters ***www.instagram.com*** into the browser.
2. The browser sends an **HTTPS GET request** to Instagram's web server.
3. The server processes the request and sends the required webpage.
4. The browser receives the response and displays the Instagram homepage.

HTTP Request contains:

- Request Line
- Header Lines
- Optional Body

HTTP Response contains:

- Status Line
- Header Lines
- Body

4. Explain How TCP Provides Reliable Communication for HTTP TCP provides:

- **Reliable Delivery:** Ensures that every data segment reaches the destination successfully.
- **Error Control:** Detects lost or damaged segments and retransmits them.
- **Flow Control:** Prevents the sender from transmitting data faster than the receiver can process it.
- **Congestion Control:** Reduces network congestion by controlling the transmission rate.

Because of TCP, Instagram pages, images, reels, and messages are delivered accurately without missing information.

5.Identify Whether the Application Follows Client–Server or Peer-to-Peer Architecture and Justify Your Answer

Instagram follows the **Client–Server architecture**.

In this architecture, the web browser or Instagram mobile application acts as the **client**, while Instagram's servers act as the **server**.

The client sends HTTPS requests to Instagram's server whenever a user logs in, views posts, watches reels, or sends messages. The server processes these requests and returns the required information to the client.

This architecture is suitable because Instagram stores user accounts, photos, videos, messages, and other data on centralized servers while providing services to millions of users simultaneously.

6.Compare HTTP and FTP

HTTP	FTP
Used by Instagram to load the website, login page, user profiles, posts, reels, and stories.	Used to transfer files between computers or servers.
Uses a single TCP connection for communication.	Uses two TCP connections (control and data).
Uses methods such as **GET** and **POST** to request webpages and submit login information or comments.	Uses commands such as **RETR, STOR, LIST,** and **DELE** for file operations.
Transfers web content such as HTML, CSS, JavaScript, images, and videos displayed on Instagram.	Transfers files of various formats between local and remote systems.

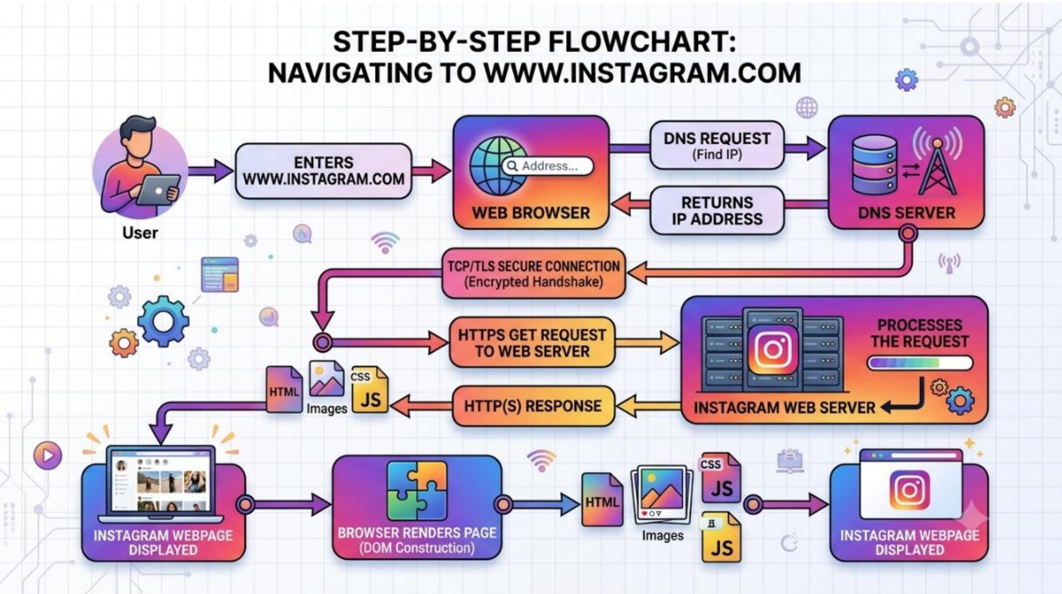
7. Explain How Cookies Improve User Experience with One Real-World Example

Cookies are small files that store information about a user in the browser. When a user logs into **Instagram**, the server sends a cookie to the browser. The browser saves this cookie and sends it back whenever the user visits Instagram again.

This helps Instagram recognize the user and keep them logged in without asking them to enter their username and password every time. Cookies also help remember user preferences, making the browsing experience faster and more convenient.

Example:
When you log in to Instagram today, you usually remain logged in even after closing and reopening the browser. This is possible because the browser stores a cookie that identifies your login session.

8. Flow diagram showing the complete communication from the browser to the web server and back.



9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Application-layer protocols are essential for accessing websites and online services like Instagram. When a user opens Instagram, **DNS** converts the website name into an IP address, **HTTPS** securely transfers data between the browser and Instagram's server, and **TCP** ensures that photos, videos, messages, and other content are delivered accurately and in the correct order. **Cookies** improve the user experience by remembering login sessions and user preferences. Working together with the TCP/IP protocol stack, these protocols provide fast, reliable, and secure communication over the Internet. Understanding these protocols helps us learn how Instagram and other modern web applications function efficiently in our daily lives.